Izmir Institute of Technology

Electrical and Electronics Eng. Dep.

EE 471 Modern Software Development Practices and Technologies

In-class exercise Week #6

Thursday 01/04/2026

Preliminary Task: Study the shared course materials [1, 2, 3,4, 5,6]

Task: Leverage vibe coding to create a speech-studio backed website

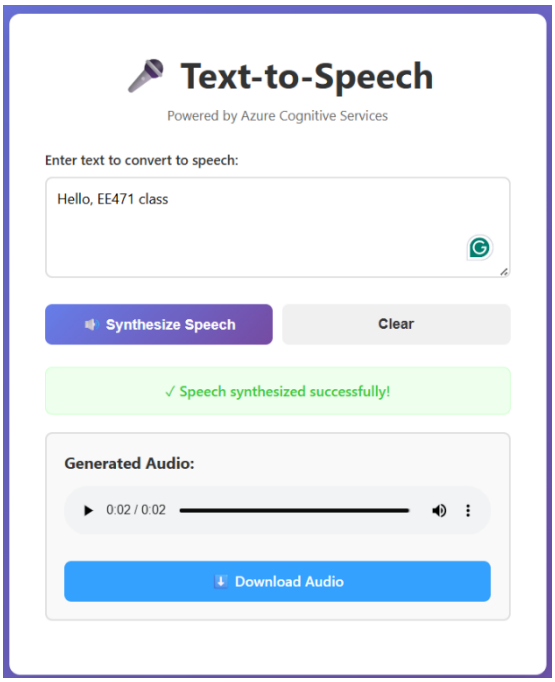
You are working in a startup, and your customer asked your team to develop a website that a user can translate:

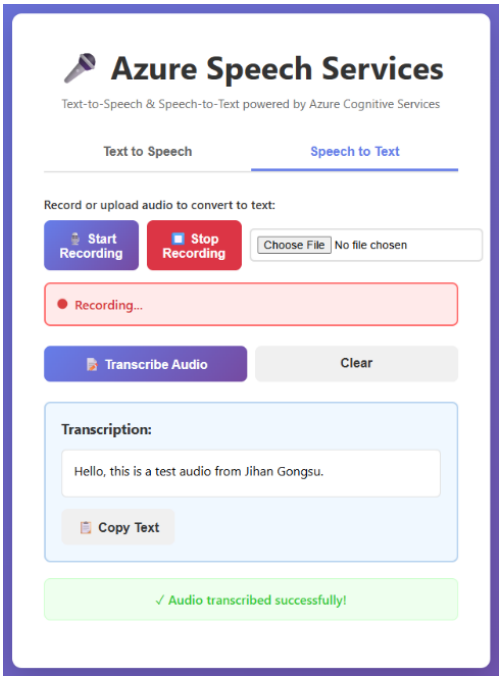
- text to speech and
- speech to text

Your team lead wants to deliver it ASAP and you decided work with a colleague of yours. As a team, one developer works on text to speech and the other is on speech to text.

PART 0 (STRONGLY RECOMMENDED):

Create a .env file to keep your secret keys, “e.g. Azure SPEECH_KEY”, and use “python-dotenv” library to use it in your scripts. Update your “.gitignore” so that “.env” file is not pushed to remote and stays in your local **as they are sensitive information**.





Part 1: Text to speech

1) In your github repository open a branch with the name "text-to-speech"
2) Develop a website running on your localhost (with a Flask-based backend). This should be capable of translating texts entered by the users to an audio file.
3) Push it to your branch, and merge it to main.

Part 2: Speech to text

1) In your github repository open a branch with the name "speech-to-text"
2) Develop a website running on your localhost (with a Flask-based backend). This should be capable of translating audio files / or speech inputs given by the users to a text script.
3) Push it to your branch, and merge it to main.

Part 3: Resolve conflicts / Update your script for both functions

1) In Part 2, merging will fail and raises merge conflicts.
2) Resolve the conflicts, and update your code so that your new website has both of the features: text-to-speech & speech-to-text

[Optional] Part 4: Extend with the rest of the supplementary materials

1) Improve your script with real-time diarization, speech translation, and pronounciation-assessment

Deliverable: Your github repo.